

Land Acquisition Process — Domain Research

A stage-by-stage reference on India's land acquisition lifecycle, aligned with the SIH problem statement cluster on Land Governance

[Purpose and Scope](#)

[Stage 1: Preliminary Notification \(Section 11\)](#)

[Stage 2: Social Impact Assessment \(SIA\)](#)

[Stage 3: Appraisal by an Independent Expert Group](#)

[Stage 4: Consent Requirement \(conditional — private and PPP projects only\)](#)

[Stage 5: Declaration \(Section 19\)](#)

[Stage 6: Award — Compensation Determination \(Sections 25–30\)](#)

[Stage 7: Compensation Disbursement](#)

[Stage 8: Possession](#)

[Stage 9: Rehabilitation & Resettlement \(R&R\)](#)

[Acquisition Routes Beyond RFCTLARR](#)

[Cross-Reference: How the Problem Statement Cluster Connects](#)

[Open Questions for Scoping](#)

Purpose and Scope

This document explains how land acquisition actually works in India today — the legal process, the stakeholders involved, and where the friction points are — before any system design begins. It draws on:

- The **RFCTLARR Act, 2013** (Right to Fair Compensation and Transparency in Land Acquisition, Rehabilitation and Resettlement Act), the central law governing government land acquisition.
- The problem statement cluster issued by the Ministry of Rural Development, Department of Land Resources, covering: the National Land Acquisition & Management System, a Predictive Analytics System for delay detection, an Intelligent Land Record Digitization system, and a National Research & Policy Platform.

The nine stages below represent the full lifecycle from the moment land is identified through final possession and rehabilitation.

Stage 1: Preliminary Notification (Section 11)

What happens: A “requiring body” — the government department or agency that needs the land (for a highway, railway line, industrial corridor, irrigation project, etc.) — identifies specific land. The appropriate acquiring authority then publishes a notification declaring the *intent* to acquire that land.

Where it is published: Three separate, disconnected channels — 1. The Official Gazette 2. Local newspapers 3. Village notice boards (panchayat/municipal offices)

This fragmentation across three unlinked publication channels is itself one of the core problems the land acquisition digitization brief calls out: no unified national platform, and inconsistent data collection across mediums.

Who is involved:

- The requiring body (initiates the request)
- The Land Acquisition Authority / District Collector's office (issues the notification)
- Affected landowners (notified, and entitled to object)

What landowners can do: File objections within a stipulated time window.

Key data points at this stage: project ID, requiring body/ministry, land parcel(s) identified (by survey number), notification date, proof of publication (gazette reference, newspaper clipping, notice-board photo), objection-window deadline.

Stage 2: Social Impact Assessment (SIA)

What happens: A mandatory study (with narrow emergency exemptions) assessing how the acquisition will affect displaced communities, livelihoods, and the local environment. It is conducted **in consultation with local Gram Sabha, Panchayat, or Municipal bodies** — not a desk exercise done in isolation.

Includes: A public hearing where affected families can raise objections and suggestions, which the government must record and consider.

A specific legal safeguard: The Act restricts acquisition of multi-crop irrigated agricultural land except in exceptional, justified circumstances — a food-security protection.

Who is involved: State government (commissions the SIA), local self-government bodies, affected families, and an independent expert group (see Stage 3).

Key data points at this stage: SIA report (document), public hearing log (attendees, objections raised, responses given), land classification flag (multi-crop irrigated or not).

Stage 3: Appraisal by an Independent Expert Group

What happens: An independent expert group reviews the SIA report. Critically, it **can recommend cancelling the project outright** if the social cost is judged to outweigh the public benefit. This is a genuine go/no-go decision point, not a rubber stamp.

Outcome states: Approved as-is / Approved with modifications / Rejected.

Key data points at this stage: Review outcome, expert group composition, justification/reasoning document.

Stage 4: Consent Requirement (conditional — private and PPP projects only)

What happens: This stage applies **only** when land is being acquired on behalf of a private company or for a Public-Private Partnership (PPP) project — not for purely government public-purpose projects.

- Private company projects: consent required from **80%** of affected families.
- PPP projects: consent required from **70%** of affected families.

Key data points at this stage: Project type (Government / Private / PPP), number of affected families, consents collected to date, live consent percentage against the applicable threshold.

Stage 5: Declaration (Section 19)

What happens: A formal declaration is published confirming the land identified will be acquired. At this point:

- The **Collector conducts a survey and physical measurement of the land**.
- The Collector **invites claims** from affected persons regarding their ownership interest, the compensation amount they believe is due, and how compensation should be apportioned among co-owners if the land is jointly held.

Why this matters for system design: This is where geo-tagging and spatial data genuinely enter the process for the first time — the survey/measurement step is a natural point to capture parcel boundaries and coordinates rather than treating GIS as a bolt-on feature.

Key data points at this stage: Parcel boundary/geo-coordinates, survey measurement record, list of ownership claims (a parcel may have multiple claimants), co-ownership apportionment logic.

Stage 6: Award — Compensation Determination (Sections 25-30)

Statutory deadline: The Collector **must announce the award within 12 months** of the declaration (Stage 5) being published. Missing this deadline without a formal extension can cause the entire proceeding to lapse — a hard,

trackable milestone.

How the compensation amount is actually calculated:

Market value (Section 26) — the **highest** of: 1. The minimum land value under the Indian Stamp Act (the official circle rate), or 2. The average of the **top 50%** of registered sale deed values for comparable land in the nearest village/area over the preceding 3 years, or 3. The amount already agreed upon in a consent award, if one exists.

Plus (Section 27): The assessed value of standing crops, trees, buildings, and other immovable assets on the land, added separately.

Plus — Solatium: A **100% solatium** — an additional premium specifically because this is a *compulsory* acquisition, not a voluntary sale.

Plus — Statutory multiplier: The final amount is scaled up to **4× market value in rural areas** and **2× market value in urban areas** — the headline “fair compensation” guarantee the Act is built around.

Plus — Interest for delay (Section 30(3)): **12% per annum** on market value, calculated from the SIA notification date (Stage 2) until the award date or possession date, whichever comes first. If payment is delayed even after the award, an additional **9% per annum** accrues from the award date until the money is actually paid.

Dispute path: A landowner who believes the award is inadequate has a **formal right to challenge it**. Disputes go to Land Acquisition and Rehabilitation Tribunals; appeals from there go to the District Collector, and further appeals go to the High Court.

Direct link to the delay-prediction problem statement: the parameters explicitly named there for predicting delays — compensation status, approval timelines, legal disputes, possession status, stakeholder responsiveness — map almost one-to-one onto the fields this stage naturally produces (award date vs. 12-month deadline, dispute filed y/n, payment date vs. award date).

Key data points at this stage: Circle rate, comparable sale deed values, asset valuation, area classification (rural/urban), notification date, award date, calculated compensation breakdown (market value + assets + solatium + multiplier + interest), dispute/appeal status.

Stage 7: Compensation Disbursement

What happens: Actual payment is made to landowners and other claimants, apportioned per the ownership claims recorded in Stage 5.

Why this stage matters beyond bookkeeping: Every day payment is delayed past the award date adds 9% per annum interest (Stage 6) — meaning payment delay has a real, quantifiable cost to the exchequer, not just an administrative inconvenience. This is a genuinely meaningful metric for a monitoring dashboard, distinct from a generic “days pending” counter.

Key data points at this stage: Disbursement status per claimant, disbursement date, running interest-accrual calculation, linkage back to the award amount.

Stage 8: Possession

What happens: Physical possession of the land is formally handed over to the requiring body.

Critical statutory trigger — Section 24(2): If compensation has **not been paid within 5 years** of the award, the acquisition proceedings can **lapse entirely**, subject to some case-law nuance on what legally counts as “paid.” A lapsed acquisition effectively restarts the entire process from Stage 1.

Direct link to the land acquisition management problem statement: this maps directly onto the “possession status” and “timeline monitoring and milestone tracking” parameters it explicitly asks for.

Key data points at this stage: Possession date, 5-year lapse-risk countdown per project, lapse status flag.

Stage 9: Rehabilitation & Resettlement (R&R)

This stage runs partly in parallel with Stages 5 through 8, rather than strictly after them.

What happens: Goes beyond monetary compensation. Includes: - Resettlement sites and housing - Transport and resettlement allowances - Employment or annuity support intended to restore livelihoods, not merely replace land value

Who is eligible — a detail that is easy to miss: R&R eligibility is **not limited to landowners**. Agricultural laborers who worked the land without owning it, and other livelihood-dependent families, can also qualify for R&R support.

Additional protections: The Act contains extra safeguards for Scheduled Castes (SC) and Scheduled Tribes (ST) communities, particularly in Scheduled Areas, in recognition of deeper social and cultural dependence on land.

Direct link to the problem statements: this is exactly why “Rehabilitation and Resettlement (R&R) progress” and “Number of affected and displaced families” are listed as their own separate tracked parameters, distinct from land and compensation data — an owner, a co-owner, and a landless agricultural laborer are three different eligibility categories that a pure “landowner” data model would fail to represent.

Key data points at this stage: Affected person record (owner / co-owner / agricultural laborer / other livelihood-dependent), SC/ST status flag where applicable, resettlement site assignment, allowance/annuity disbursement status.

Acquisition Routes Beyond RFCTLARR

Not all land acquisition in India happens under RFCTLARR. Two other routes exist and are worth acknowledging even where only RFCTLARR is modeled in depth:

- Sector-specific Acts** — for example, the Railway Act (1989, amended 2008) is explicitly exempted from RFCTLARR under its Schedule IV. Railways can acquire land under their own Act, which follows a different (often faster) procedure.
- Private negotiated purchase** — private developers may simply buy land directly through standard Sale Deeds without invoking any compulsory acquisition power. This is not “land acquisition” in the legal sense used by RFCTLARR.

A system describing itself as a *national* platform should be able to at least tag which route a given project falls under, even if only the RFCTLARR route is fully modeled end-to-end.

Cross-Reference: How the Problem Statement Cluster Connects

The Ministry of Rural Development issued four related problem statements that all draw from this same underlying process:

Problem Statement	What it asks for	Which stage(s) it draws from
National Land Acquisition & Management System	End-to-end digital workflow: proposal → approval → notification → award → compensation → possession; GIS geo-tagging; national dashboard; MIS reports	All 9 stages — this is the umbrella system
Predictive Analytics for Delay Detection	Risk-score each project, predict delay probability using project type, land area, affected families, compensation status, approval timelines, legal disputes, possession status, rehabilitation progress, stakeholder responsiveness	Primarily Stages 6 and 8 (award deadline, dispute status, 5-year lapse risk)
Intelligent Land Record Digitization	OCR/NLP extraction of landowner details, survey number, khasra number, khata number, plot area, village, tehsil, district, land classification, ownership details, mutation records, registration information from scanned/handwritten legacy records	Primarily Stage 5 (ownership claims, survey and measurement) and Stage 1 (parcel identification)
National Research & Policy Platform	Centralized repository of research, policy documents, and datasets for evidence-based land governance; policy simulation	Cuts across all stages as an analytics/knowledge layer, not a transactional one

Two field-naming details worth carrying forward precisely, since they come directly from the digitization problem

statement and are the actual government terminology for land records: **khasra number** (a plot identification number used in rural land records) and **khata number** (a landowner account/holding number that can group multiple khasra plots under one owner). These are more specific and accurate than a generic “survey number” field, and should be reflected in any data model that touches ownership records.

Open Questions for Scoping

These are genuine decisions to make before finalizing a data model — not busywork:

1. Which acquisition route(s) get modeled — RFCTLARR only, or RFCTLARR plus a lighter Railway Act variant?
2. Is “Affected Person” modeled as an entity distinct from “Landowner” (recommended, given Stage 9’s laborer-eligibility detail)?
3. Is the dispute/appeal sub-workflow from Stage 6 modeled, or noted as future scope?
4. Is consent tracking (Stage 4) modeled, or skipped on the basis that a government public-purpose project narrative (which skips this stage entirely) is being used for the demo?